## Contact

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## Top Skills

TensorFlow
Amazon Web Services (AWS)
JavaScript

# Tyler Bednar

Staff AI/ML Engineer | Forward Deployed Engineer | LLM Fine-Tuning & Real-Time Inference Pipelines | PyTorch, LangGraph
United States

## Summary

AI/ML Engineer with 10+ years of experience designing, building, and deploying intelligent systems across the full machine learning lifecycle — from research prototyping to production-grade inference at scale.Started my career in data science and statistical modeling, building predictive analytics pipelines, recommendation engines, and time-series forecasting systems using scikit-learn, XGBoost, and Spark MLlib. Spent the early years deep in feature engineering, A/B testing frameworks, and building ETL workflows that fed ML models serving millions of requests daily.Transitioned into deep learning and NLP, architecting custom model training pipelines with PyTorch and TensorFlow for text classification, named entity recognition, sentiment analysis, and document intelligence. Built end-to-end computer vision solutions for object detection and image segmentation using YOLO, ResNet, and EfficientNet, deployed on edge devices and cloud endpoints.Over the last few years, shifted focus heavily into LLMs and Generative AI — designing production RAG architectures with LangChain, LlamaIndex, and vector databases like Pinecone, Weaviate, and ChromaDB. Built multi-agent orchestration systems using LangGraph, fine-tuned open-source models (LLaMA, Mistral, Falcon) with LoRA/QLoRA and RLHF, and optimized inference with vLLM, ONNX, and Triton Inference Server.On the infrastructure side, I've owned MLOps pipelines end-to-end — model versioning with MLflow and DVC, distributed training with DeepSpeed and Ray, containerized deployments on Kubernetes, and CI/CD for model serving using GitHub Actions, Airflow, and Terraform. Comfortable operating across AWS (SageMaker, Bedrock, EKS, Lambda), Azure (Azure ML, OpenAI Service), and GCP (Vertex AI, BigQuery ML, TPU pods).Core technical stack spans Python, SQL, Spark, FastAPI, Docker, and infrastructure-as-code tooling. Strong in experiment tracking (Weights & Biases, Neptune), data orchestration (Airflow, Prefect, dbt), and monitoring/observability for production models (Evidently, Prometheus, Grafana).What drives me is

taking ML from notebook to production — closing the gap between research and real-world impact. Whether it's reducing inference latency by 60%, building a RAG pipeline that cuts hallucination rates, or standing up an evaluation framework for LLM outputs, I care about systems that actually work at scale.

## Experience

Johnson & Johnson
Staff AI/ML Engineer/FDE
August 2022 - Present (4 years 3 months)

Wells Fargo
Staff AI Engineer
July 2018 - August 2022 (4 years 2 months)

The Home Depot
Machine Learning Engineer
November 2016 - July 2018 (1 year 9 months)

## Education

Farmingdale State College
Bachelor's Degree, Computer and Information Systems Security/Information Assurance · (2010 - 2014)

The King's College
Master's Degree · (2014 - 2016)